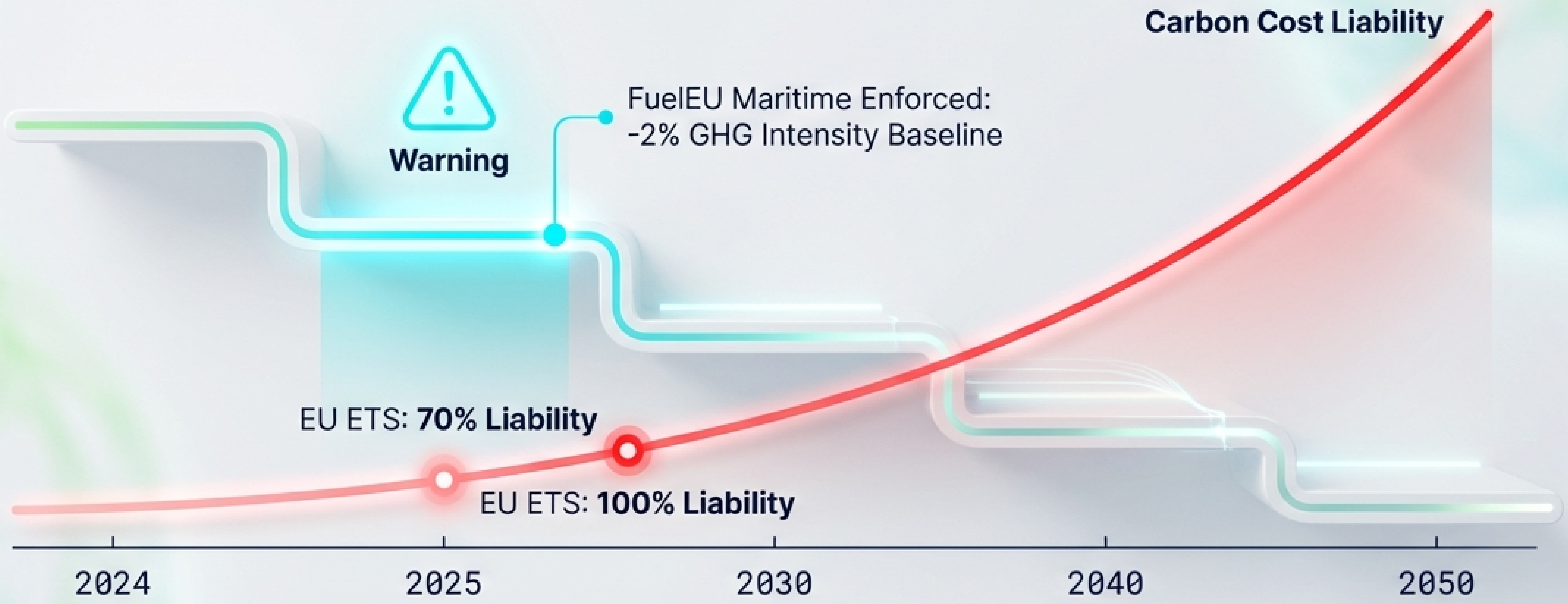
Navigating the Green Transition

Operational Strategies for Biofuel Integration & Lubrication Excellence

Strategic Analysis 2025-2030

2025 is the Regulatory Tipping Point



IMO CII Rating
Annual tightening of ~2%

FuelEU Baseline
91.16 gCO2e/MJ

The Mandate
Negative Exponential Curve to Net Zero

Decarbonization is Now a Financial Metric

Stranded Assets

Ships rated CII ‘D’ or ‘E’ risk commercial obsolescence and mandatory corrective plans

FuelEU Penalties apply per non-compliant MJ

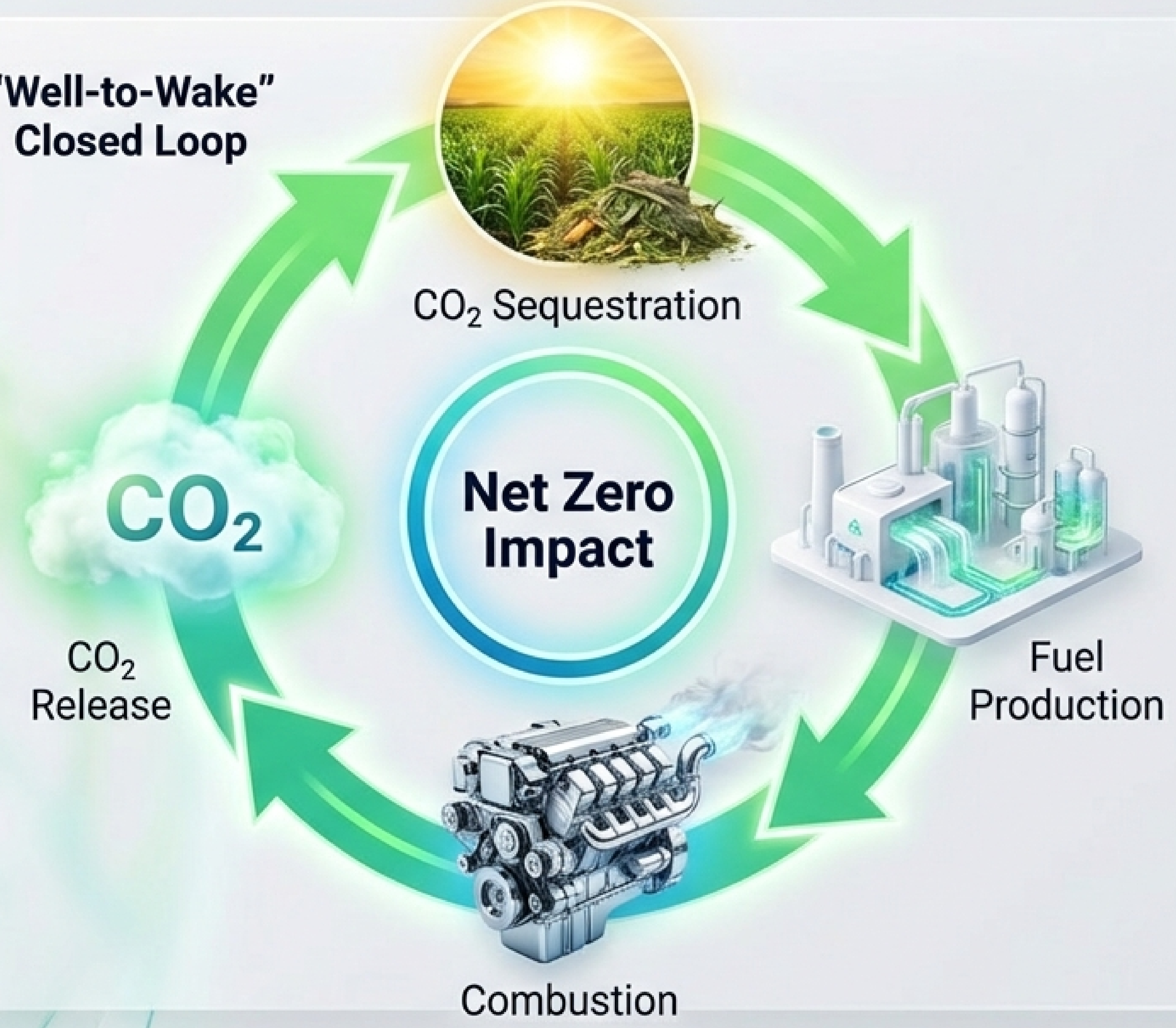
Scope 3 Demands

Major retailers (IKEA, Amazon) require ‘Green Shipping Services’ to meet their own supply chain targets

EU ETS Savings offset Biofuel premiums

The Only "Drop-In" Bridge to 2030

"Well-to-Wake" Closed Loop



Zero CAPEX: No hardware retrofits required

Immediate Impact: B25 blend moves CII Rating **D → C**

Low WtT Emissions: Waste-based feedstocks offer negative upstream emissions

Know Your Molecules: FAME vs. HVO

FAME (Biodiesel)

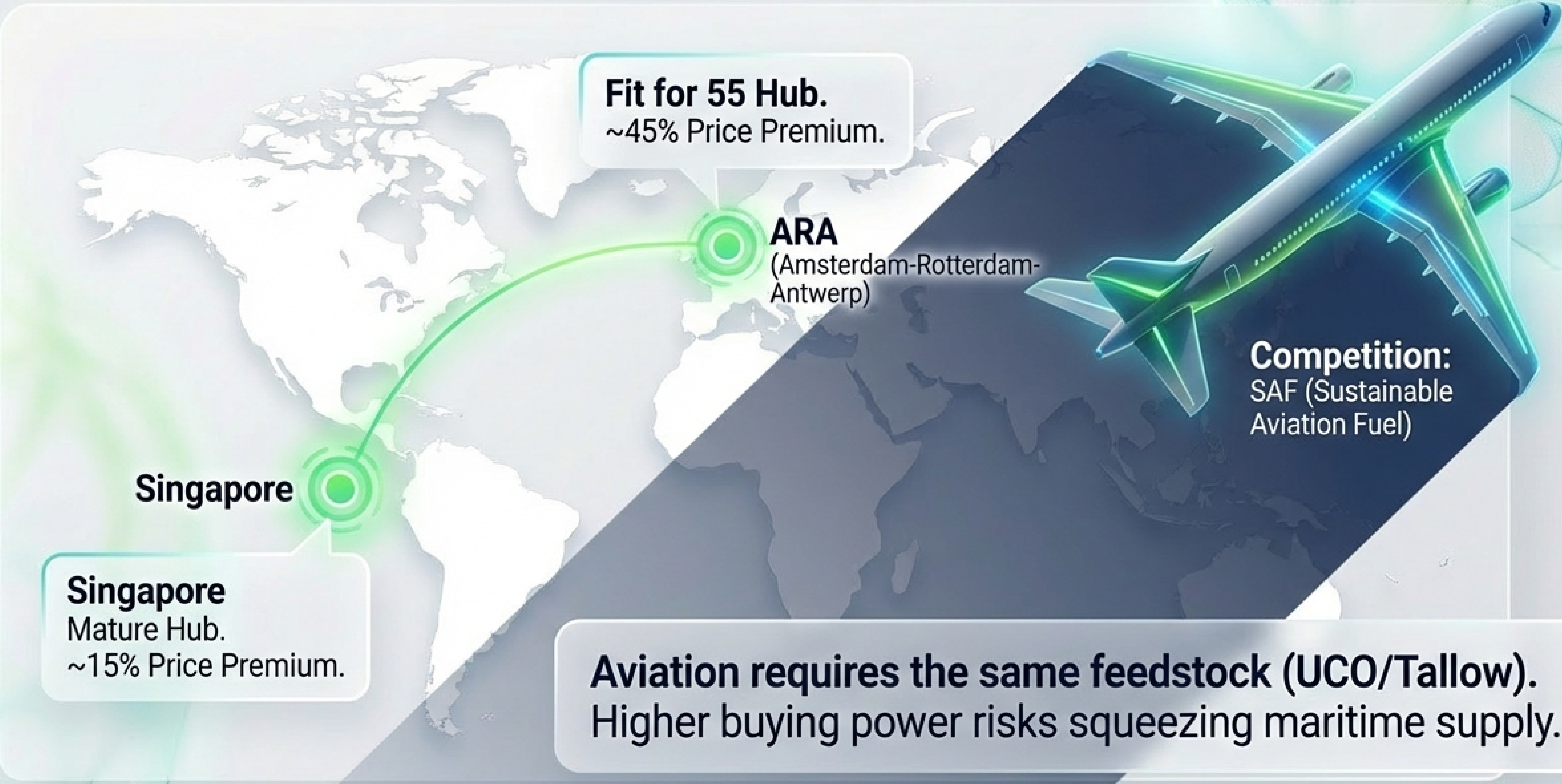
- **Generation:** 1st Gen (Esterification)
- **Chemistry:** ~10% Oxygen Content
- **Properties:** Hydrophilic (Water loving), Solvent Effect, Lower Heating Value.

HVO (Renewable Diesel)

- **Generation:** 2nd Gen (Hydro-processed)
- **Chemistry:** Paraffinic (No Oxygen, No Aromatics)
- **Properties:** High Cetane, Cold Resistant, Identical to Fossil Diesel.

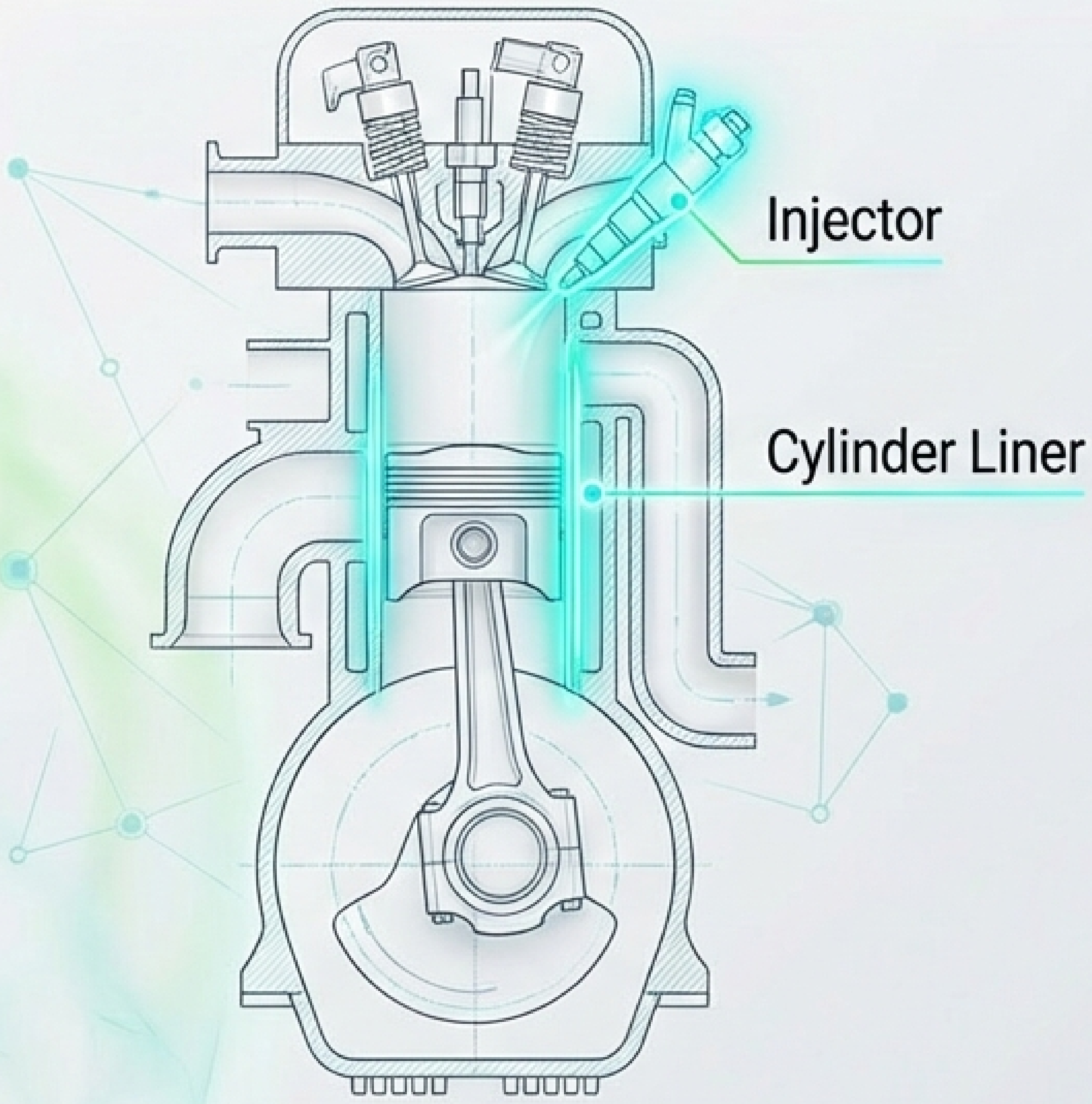
HVO is the superior technical drop-in, while FAME is the cost-effective market standard.

Global Supply Chain and The Aviation Threat



NotebookLM

Engine Readiness and Hardware Vigilance



MAN ES & WinGD Status:
Fully Compatible.

Risk 1: Injector Wear.
Lower calorific value requires longer injection times, accelerating component fatigue.

Risk 2: The Solvent Effect.
FAME cleans sludge from tanks, risking sudden filter clogging.

Risk 3: Low Load Operations.
Slow steaming requires vigilance to prevent fuel dilution.

The Hidden Challenge: Cylinder Lubrication Risks

Intact Oil Film

Film Breakdown & Contact

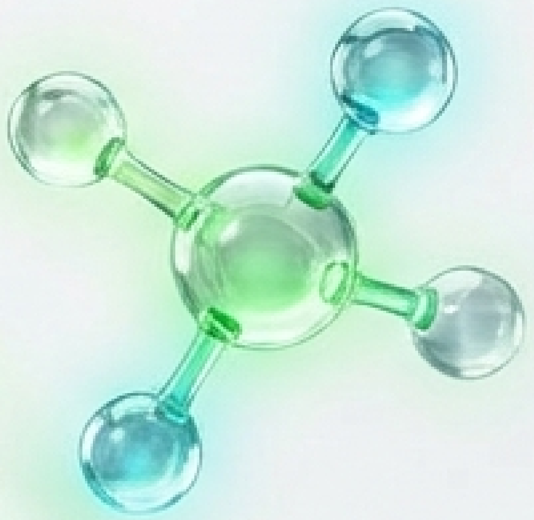
Viscosity Drop
Biofuel dilution thins the oil film, reducing hydrodynamic protection.

The Cleanliness Paradox
Low sulfur means low acid, but high cleanliness is still required. Standard Low-BN oils may lack sufficient detergency.

Corrosive Indicators
Scrape-down analysis often shows spikes in Iron (Fe) and Molybdenum (Mo).

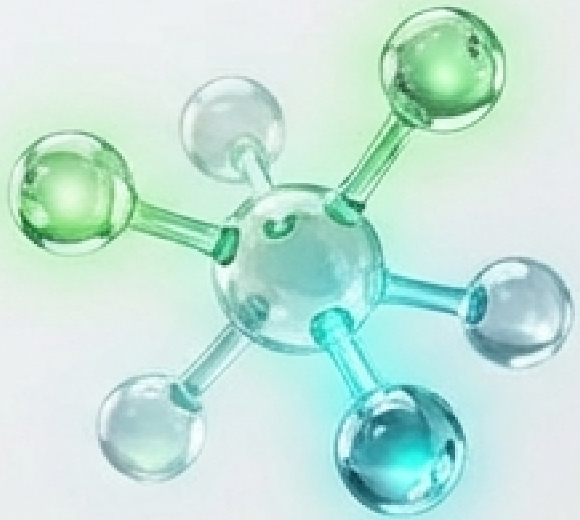
Next-Generation Lubrication Strategies

Adapting Chemistry for Bio-Integration



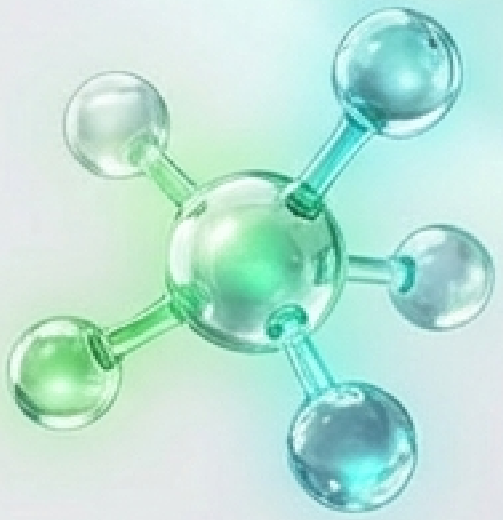
Advanced Category II Formulations

High-performance dispersion chemistry (BN 40). Decouples alkalinity from cleanliness to handle deposits without ash buildup.



Ashless Neutralizing Molecules

Non-metallic acid neutralization. Neutralizes combustion byproducts without calcium-based detergents, preventing crown deposits.



Balanced Anti-Scuffing Chemistry

Enhanced thermal stability. Boosted anti-wear agents maintain the boundary layer even during viscosity loss.

Operational Best Practices: The First 48 Hours

Chief Engineer’s Dashboard

Filtration Alert

ΔP: RISING

Monitor differential pressure closely. FAME solvent effect strips strips tank sludge immediately.

Water Management

Water Management

Biofuels are hydrophilic. Maintain strict tank dryness to prevent microbial growth.

H_2O CONTENT: <0.05% REQUIRED

Thermal Control

Thermal Control

Adjust service temps. HVO and FAME require different injection viscosities than VLSFO.

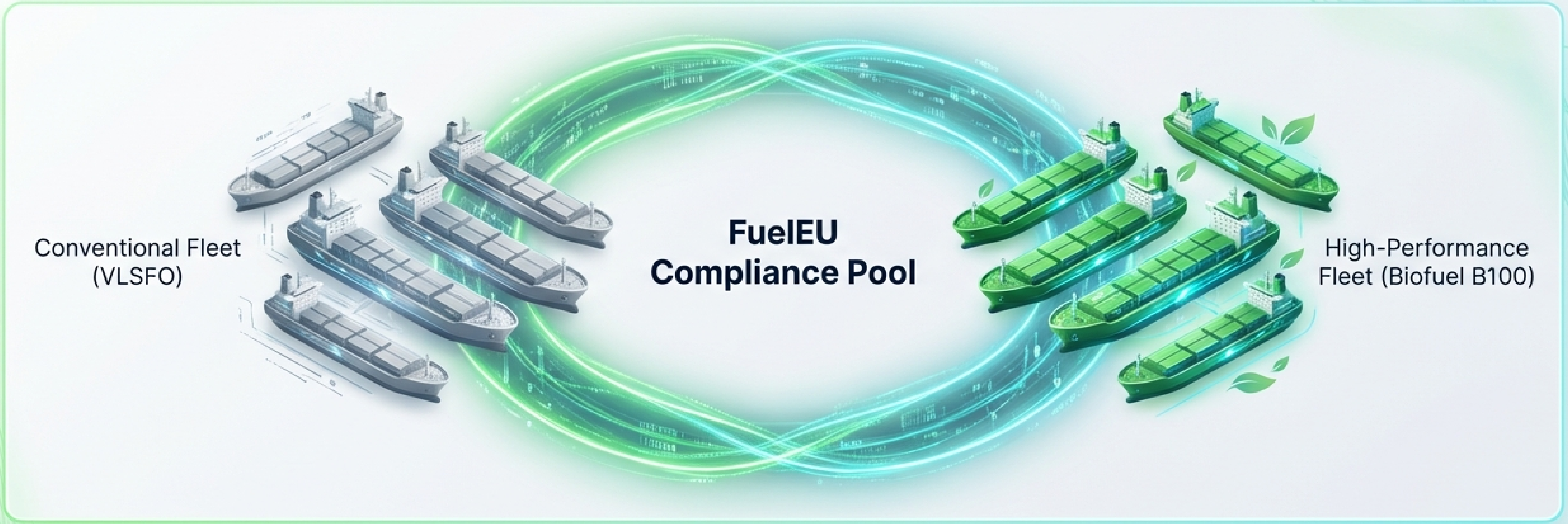
TEMP SETTING: OPTIMIZING FOR VISCOSITY

Condition Monitoring

Condition Monitoring

Frequent Scrape-Down Analysis. Watch for Fe spikes or sudden BN depletion.

Strategic Fleet Management & Pooling



Pooling Mechanism

Surplus compliance from 'Green' ships offsets deficits of 'Grey' ships.

Dynamic Usage

Deploy Biofuel tactically when EU ETS prices spike.

Rating Defense

Targeted use to prevent specific vessels from dropping to CII Rating D/E.

The Green Bridge to Net Zero

Biofuel is the **critical bridge** that allows **decarbonization** *today* while future **technology** matures.

References & Technical Sources

PRIMARY SOURCES

- MAN Energy Solutions Service Letters (SL2023-741)
  Roboto Mono
- WinGD General Technical Data (GTD)
  Roboto Mono
- IMO & EU Regulatory Documents (FuelEU Maritime, CII)
  Roboto Mono

INDUSTRY & RESEARCH

- Industry Reports: Argus Media, Lloyd's Register, DNV
  Roboto Mono
- Advanced Lubrication Research (Category II & ANM Technologies)
  Roboto Mono
- USDA Foreign Agricultural Service Biofuels Annual
  Roboto Mono